

Shopping Mission Interpreter — Evaluation Report

50 test examples · 3 tiers (Easy 16 / Medium 18 / Hard 16) · Systems: keyword, llm

≥ 0.85 (good) 0.50 – 0.85 (moderate) < 0.50 (weak) **+0.05+ delta = LLM wins** **~0 delta = tie**

System: Keyword Baseline

Field	Metric	ALL (50)	EASY (16)	MEDIUM (18)	HARD (16)
meal_occasion	Accuracy	0.940	0.875	1.000	0.938
mission_type	Accuracy	0.460	0.000	0.611	0.750
budget_sensitivity	Accuracy	1.000	1.000	1.000	1.000
List fields — Micro F1					
dietary_constraints	F1	0.857	0.000	1.000	0.769
preference_constraints	F1	0.772	0.000	0.800	0.833
product_categories	F1	0.514	0.952	0.205	0.286
risk_flags	F1	0.450	0.000	0.222	0.516
excluded_items	F1	0.750	0.000	0.000	0.750

System: LLM (gpt-4o-mini)

Field	Metric	ALL (50)	EASY (16)	MEDIUM (18)	HARD (16)
meal_occasion	Accuracy	0.960	0.938	1.000	0.938
mission_type	Accuracy	0.620	0.500	0.778	0.562
budget_sensitivity	Accuracy	1.000	1.000	1.000	1.000
List fields — Micro F1					
dietary_constraints	F1	0.957	0.000	1.000	0.933
preference_constraints	F1	0.786	0.000	0.788	0.818
product_categories	F1	0.618	0.976	0.407	0.500
risk_flags	F1	0.711	0.000	0.182	0.882
excluded_items	F1	1.000	0.000	0.000	1.000

Keyword vs LLM — Side by Side (All Tiers)

Field	Metric	Keyword	LLM	Delta (LLM – KW)
meal_occasion	Accuracy	0.940	0.960	+0.020
mission_type	Accuracy	0.460	0.620	+0.160
budget_sensitivity	Accuracy	1.000	1.000	+0.000

Field	Metric	Keyword	LLM	Delta (LLM – KW)
List fields — Micro F1				
dietary_constraints	F1	0.857	0.957	+0.099
preference_constraints	F1	0.772	0.786	+0.014
product_categories	F1	0.514	0.618	+0.104
risk_flags	F1	0.450	0.711	+0.261
excluded_items	F1	0.750	1.000	+0.250

Failure Analysis — Keyword Baseline

49 / 50 examples had at least one incorrect field | Easy: 16 Medium: 18 Hard: 15

ID	Tier	Request	Failed Fields	Key difference
e01	easy	Milk, bread and eggs please	mission_type	mission_type: gold= <i>household_stock_up</i> pred= <i>unclear</i>
e02	easy	I need cereal and coffee for breakfast	mission_type, preference_constraints	mission_type: gold= <i>specific_ingredient</i> pred= <i>convenience_meal</i> preference_constraints: gold= <i>∅</i> pred= <i>quick</i>
e03	easy	I want to buy bananas, apples and oranges	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e04	easy	Chicken breast and rice	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e05	easy	Get me some yogurt	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e06	easy	Pasta and tomato sauce for dinner	mission_type, product_categories	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i> product_categories: gold= <i>pantry_staples</i> pred= <i>produce, pantry_staples</i>
e07	easy	Two bottles of red wine	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e08	easy	Dish soap and laundry detergent	mission_type	mission_type: gold= <i>household_stock_up</i> pred= <i>unclear</i>
e09	easy	I need diapers and baby formula	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e10	easy	Some cheese and crackers	meal_occasion, mission_type	meal_occasion: gold= <i>snack</i> pred= <i>none</i> mission_type:

ID	Tier	Request	Failed Fields	Key difference
				gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e11	easy	Orange juice and toast for breakfast	mission_type, preference_constraints, product_categories	mission_type: gold= <i>specific_ingredient</i> pred= <i>convenience_meal</i> preference_constraints: gold= $\emptyset$ pred= <i>quick</i> product_categories: gold= <i>beverages, bakery</i> pred= <i>produce, bakery, beverages</i>
e12	easy	A bag of frozen peas	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e13	easy	Ground beef for burgers	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e14	easy	Olive oil and salt	mission_type	mission_type: gold= <i>household_stock_up</i> pred= <i>unclear</i>
e15	easy	Chocolate bars and chips	meal_occasion, mission_type, preference_constraints	meal_occasion: gold= <i>snack</i> pred= <i>none</i> mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i> preference_constraints: gold= <i>indulgent</i> pred= $\emptyset$
e16	easy	Sparkling water, six bottles	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
m01	medium	Something quick and easy for dinner tonight	product_categories	product_categories: gold= <i>ready_meals, frozen</i> pred= $\emptyset$
m02	medium	Healthy lunch options for the office	product_categories	product_categories: gold= <i>produce, ready_meals</i> pred= $\emptyset$
m03	medium	I'm hosting friends this weekend, need party food	product_categories	product_categories: gold= <i>snacks_sweets, alcohol, beverages</i> pred= $\emptyset$
m04	medium	Snacks the kids will like for after school	product_categories	product_categories: gold= <i>snacks_sweets, dairy_eggs</i> pred= $\emptyset$

ID	Tier	Request	Failed Fields	Key difference
m05	medium	I want to eat healthier, what should I get	mission_type, preference_constraints, product_categories, risk_flags	mission_type: gold= <i>health_goal</i> pred= <i>unclear</i> preference_constraints: gold= <i>healthy</i> pred= $\emptyset$ product_categories: gold= <i>produce, health_diet</i> pred= $\emptyset$ risk_flags: gold= <i>missing_information</i> pred= <i>ambiguous_request, missing_information</i>
m06	medium	Cheap filling meals for the week, I'm on a tight budget	product_categories	product_categories: gold= <i>pantry_staples, frozen, meat_seafood</i> pred= $\emptyset$
m07	medium	Vegetarian dinner ideas	mission_type, product_categories, risk_flags	mission_type: gold= <i>convenience_meal</i> pred= <i>unclear</i> product_categories: gold= <i>produce, ready_meals</i> pred= $\emptyset$ risk_flags: gold= $\emptyset$ pred= <i>allergy_or_safety</i>
m08	medium	Need a quick high-protein breakfast	product_categories	product_categories: gold= <i>dairy_eggs, breakfast</i> pred= $\emptyset$
m09	medium	Gluten-free bread and pasta	mission_type, risk_flags	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i> risk_flags: gold= $\emptyset$ pred= <i>negation_present, allergy_or_safety</i>
m10	medium	Premium ingredients for a special date-night dinner	mission_type, preference_constraints, product_categories	mission_type: gold= <i>party_hosting</i> pred= <i>unclear</i> preference_constraints: gold= <i>indulgent</i> pred= $\emptyset$ product_categories: gold= <i>meat_seafood, alcohol</i> pred= $\emptyset$
m11	medium	Quick snacks I can eat at my desk	preference_constraints, product_categories	preference_constraints: gold= <i>quick, easy_to_carry, no_cooking</i> pred= <i>quick, easy_to_carry</i> product_categories: gold= <i>snacks_sweets</i> pred= $\emptyset$
m12	medium	Ingredients for a kids birthday party	product_categories	product_categories: gold= <i>snacks_sweets, bakery, beverages</i> pred= $\emptyset$
m13	medium	Low-sugar options, trying to cut back	product_categories	product_categories: gold= <i>produce, health_diet</i> pred= $\emptyset$

ID	Tier	Request	Failed Fields	Key difference
m14	medium	Vegan protein sources	mission_type, product_categories, risk_flags	mission_type: gold= <i>health_goal</i> pred= <i>unclear</i> product_categories: gold= <i>health_diet</i> , <i>pantry_staples</i> pred= $\emptyset$ risk_flags: gold= $\emptyset$ pred= <i>allergy_or_safety</i>
m15	medium	Easy no-cook lunch	preference_constraints, product_categories	preference_constraints: gold= <i>no_cooking</i> , <i>quick</i> pred= <i>no_cooking</i> product_categories: gold= <i>ready_meals</i> pred= $\emptyset$
m16	medium	Halal chicken for dinner	mission_type, risk_flags	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i> risk_flags: gold= $\emptyset$ pred= <i>allergy_or_safety</i>
m17	medium	Ingredients for a green salad	mission_type, preference_constraints	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i> preference_constraints: gold= <i>healthy</i> , <i>light</i> pred= $\emptyset$
m18	medium	Stock up on basics for the month	product_categories, risk_flags	product_categories: gold= <i>pantry_staples</i> , <i>household_nonfood</i> pred= $\emptyset$ risk_flags: gold= <i>missing_information</i> pred= $\emptyset$
h01	hard	Snacks for the kids, but one of them is allergic to nuts	product_categories	product_categories: gold= <i>snacks_sweets</i> , <i>dairy_eggs</i> pred= $\emptyset$
h02	hard	Breakfast but not eggs, I can't stand them	preference_constraints, product_categories	preference_constraints: gold= $\emptyset$ pred= <i>quick</i> product_categories: gold= <i>breakfast</i> , <i>bakery</i> pred= <i>dairy_eggs</i>
h03	hard	Something healthy but I still want it to feel indulgent	product_categories, risk_flags	product_categories: gold= <i>produce</i> , <i>snacks_sweets</i> pred= $\emptyset$ risk_flags: gold= <i>conflicting_goals</i> pred= $\emptyset$
h04	hard	Quick dinner, nothing with dairy though	dietary_constraints, product_categories, risk_flags, excluded_items	dietary_constraints: gold= <i>lactose_free</i> pred= $\emptyset$ product_categories: gold= <i>ready_meals</i> pred= $\emptyset$ risk_flags: gold= <i>negation_present</i> pred= $\emptyset$ excluded_items: gold= <i>dairy</i> pred= $\emptyset$
h05	hard	Birthday cake but my daughter is celiac	dietary_constraints, risk_flags	dietary_constraints: gold= <i>gluten_free</i> pred= $\emptyset$

ID	Tier	Request	Failed Fields	Key difference
				risk_flags: gold= <i>allergy_or_safety</i> pred= $\emptyset$
h06	hard	I want to lose weight but my husband wants comfort food	product_categories, risk_flags	product_categories: gold= <i>produce, ready_meals</i> pred= $\emptyset$ risk_flags: gold= <i>conflicting_goals</i> pred= $\emptyset$
h07	hard	Dinner my vegetarian sister and my meat-loving dad will both eat	mission_type, product_categories, risk_flags	mission_type: gold= <i>party_hosting</i> pred= <i>unclear</i> product_categories: gold= <i>produce, pantry_staples</i> pred= <i>meat_seafood</i> risk_flags: gold= <i>conflicting_goals</i> pred= <i>allergy_or_safety</i>
h08	hard	Just need something, I don't know, for later	risk_flags	risk_flags: gold= <i>ambiguous_request, missing_information</i> pred= <i>negation_present, ambiguous_request, missing_information</i>
h09	hard	Protein but no meat and no soy	mission_type, dietary_constraints, product_categories	mission_type: gold= <i>health_goal</i> pred= <i>unclear</i> dietary_constraints: gold= <i>vegetarian</i> pred= $\emptyset$ product_categories: gold= <i>dairy_eggs, health_diet</i> pred= <i>meat_seafood</i>
h10	hard	Party drinks but nothing alcoholic, kids will be there	risk_flags, excluded_items	risk_flags: gold= <i>negation_present</i> pred= $\emptyset$ excluded_items: gold= <i>alcohol</i> pred= $\emptyset$
h11	hard	I'm diabetic, need a breakfast that won't spike my sugar	mission_type, preference_constraints, product_categories, risk_flags	mission_type: gold= <i>health_goal</i> pred= <i>convenience_meal</i> preference_constraints: gold= <i>low_sugar</i> pred= <i>quick</i> product_categories: gold= <i>breakfast, dairy_eggs</i> pred= $\emptyset$ risk_flags: gold= <i>allergy_or_safety</i> pred= $\emptyset$
h12	hard	Cheap but good quality wine for a dinner party	meal_occasion, risk_flags	meal_occasion: gold= <i>none</i> pred= <i>dinner</i> risk_flags: gold= <i>conflicting_goals</i> pred= $\emptyset$

ID	Tier	Request	Failed Fields	Key difference
h13	hard	Lactose-free milk and also some cheese for the kids	preference_constraints, risk_flags	preference_constraints: gold= $\emptyset$ pred= <i>kid_friendly</i> risk_flags: gold= <i>ambiguous_request</i> pred= <i>negation_present</i> , <i>allergy_or_safety</i>
h14	hard	Something filling but light, you know what I mean	risk_flags	risk_flags: gold= <i>conflicting_goals</i> , <i>ambiguous_request</i> pred= $\emptyset$
h15	hard	Vegan dinner, but my partner is allergic to nuts so be careful	mission_type, product_categories	mission_type: gold= <i>convenience_meal</i> pred= <i>unclear</i> product_categories: gold= <i>produce</i> , <i>health_diet</i> pred= $\emptyset$

Failure Analysis — LLM (gpt-4o-mini)

37 / 50 examples had at least one incorrect field | Easy: 9 Medium: 15 Hard: 13

ID	Tier	Request	Failed Fields	Key difference
e01	easy	Milk, bread and eggs please	product_categories	product_categories: gold= <i>dairy_eggs</i> , <i>bakery</i> pred= <i>dairy_eggs</i> , <i>bakery</i> , <i>pantry_staples</i>
e02	easy	I need cereal and coffee for breakfast	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>household_stock_up</i>
e06	easy	Pasta and tomato sauce for dinner	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>convenience_meal</i>
e07	easy	Two bottles of red wine	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>
e09	easy	I need diapers and baby formula	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>household_stock_up</i>
e10	easy	Some cheese and crackers	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>convenience_meal</i>
e11	easy	Orange juice and toast for breakfast	mission_type	mission_type: gold= <i>specific_ingredient</i> pred= <i>convenience_meal</i>
e14	easy	Olive oil and salt	mission_type	mission_type: gold= <i>household_stock_up</i> pred= <i>specific_ingredient</i>
e15	easy	Chocolate bars and chips	meal_occasion, mission_type, preference_constraints	meal_occasion: gold= <i>snack</i> pred= <i>none</i> mission_type: gold= <i>specific_ingredient</i> pred= <i>unclear</i>

ID	Tier	Request	Failed Fields	Key difference
				preference_constraints: gold= <i>indulgent</i> pred= $\emptyset$
m01	medium	Something quick and easy for dinner tonight	preference_constraints, product_categories, risk_flags	preference_constraints: gold= <i>quick</i> pred= <i>quick</i> , <i>easy_to_carry</i> product_categories: gold= <i>ready_meals</i> , <i>frozen</i> pred= $\emptyset$ risk_flags: gold= $\emptyset$ pred= <i>ambiguous_request</i>
m02	medium	Healthy lunch options for the office	product_categories, risk_flags	product_categories: gold= <i>produce</i> , <i>ready_meals</i> pred= $\emptyset$ risk_flags: gold= $\emptyset$ pred= <i>ambiguous_request</i>
m03	medium	I'm hosting friends this weekend, need party food	product_categories, risk_flags	product_categories: gold= <i>snacks_sweets</i> , <i>alcohol</i> , <i>beverages</i> pred= $\emptyset$ risk_flags: gold= $\emptyset$ pred= <i>ambiguous_request</i>
m04	medium	Snacks the kids will like for after school	product_categories	product_categories: gold= <i>snacks_sweets</i> , <i>dairy_eggs</i> pred= <i>snacks_sweets</i>
m05	medium	I want to eat healthier, what should I get	mission_type, product_categories	mission_type: gold= <i>health_goal</i> pred= <i>unclear</i> product_categories: gold= <i>produce</i> , <i>health_diet</i> pred= $\emptyset$
m06	medium	Cheap filling meals for the week, I'm on a tight budget	product_categories	product_categories: gold= <i>pantry_staples</i> , <i>frozen</i> , <i>meat_seafood</i> pred= $\emptyset$
m07	medium	Vegetarian dinner ideas	mission_type, product_categories, risk_flags	mission_type: gold= <i>convenience_meal</i> pred= <i>unclear</i> product_categories: gold= <i>produce</i> , <i>ready_meals</i> pred= $\emptyset$ risk_flags: gold= $\emptyset$ pred= <i>ambiguous_request</i>
m08	medium	Need a quick high-protein breakfast	product_categories	product_categories: gold= <i>dairy_eggs</i> , <i>breakfast</i> pred= <i>breakfast</i>
m10	medium	Premium ingredients for a special date-night dinner	mission_type, preference_constraints, product_categories	mission_type: gold= <i>party_hosting</i> pred= <i>unclear</i> preference_constraints: gold= <i>indulgent</i> pred= $\emptyset$ product_categories: gold= <i>meat_seafood</i> , <i>alcohol</i> pred= $\emptyset$
m11	medium	Quick snacks I can eat at my desk	mission_type, preference_constraints	mission_type: gold= <i>convenience_meal</i>

ID	Tier	Request	Failed Fields	Key difference
				pred=<i>unclear</i> preference_constraints: gold=<i>quick, easy_to_carry, no_cooking</i> pred=<i>quick, easy_to_carry</i>
m12	medium	Ingredients for a kids birthday party	product_categories, risk_flags	product_categories: gold=<i>snacks_sweets, bakery, beverages</i> pred=$\emptyset$ risk_flags: gold=$\emptyset$ pred=<i>ambiguous_request</i>
m13	medium	Low-sugar options, trying to cut back	product_categories	product_categories: gold=<i>produce, health_diet</i> pred=$\emptyset$
m15	medium	Easy no-cook lunch	preference_constraints, product_categories, risk_flags	preference_constraints: gold=<i>no_cooking, quick</i> pred=<i>easy_to_carry, no_cooking</i> product_categories: gold=<i>ready_meals</i> pred=$\emptyset$ risk_flags: gold=$\emptyset$ pred=<i>ambiguous_request</i>
m17	medium	Ingredients for a green salad	preference_constraints, risk_flags	preference_constraints: gold=<i>healthy, light</i> pred=$\emptyset$ risk_flags: gold=$\emptyset$ pred=<i>ambiguous_request</i>
m18	medium	Stock up on basics for the month	product_categories, risk_flags	product_categories: gold=<i>pantry_staples, household_nonfood</i> pred=<i>pantry_staples, produce, meat_seafood, dairy_eggs, bakery, frozen, snacks_sweets, ready_meals, beverages, household_nonfood</i> risk_flags: gold=<i>missing_information</i> pred=<i>ambiguous_request</i>
h01	hard	Snacks for the kids, but one of them is allergic to nuts	preference_constraints, product_categories	preference_constraints: gold=<i>kid_friendly</i> pred=$\emptyset$ product_categories: gold=<i>snacks_sweets, dairy_eggs</i> pred=<i>snacks_sweets</i>
h02	hard	Breakfast but not eggs, I can't stand them	mission_type, product_categories	mission_type: gold=<i>convenience_meal</i> pred=<i>unclear</i> product_categories: gold=<i>breakfast, bakery</i> pred=<i>breakfast</i>
h03	hard	Something healthy but I still want it to feel indulgent	mission_type, product_categories	mission_type: gold=<i>health_goal</i> pred=<i>unclear</i> product_categories: gold=<i>produce, snacks_sweets</i> pred=$\emptyset$

ID	Tier	Request	Failed Fields	Key difference
h04	hard	Quick dinner, nothing with dairy though	mission_type, product_categories	mission_type: gold= <i>convenience_meal</i> pred= <i>unclear</i> product_categories: gold= <i>ready_meals</i> pred= $\emptyset$
h06	hard	I want to lose weight but my husband wants comfort food	mission_type, preference_constraints, product_categories	mission_type: gold= <i>health_goal</i> pred= <i>unclear</i> preference_constraints: gold= <i>low_calorie, indulgent</i> pred= <i>healthy, indulgent</i> product_categories: gold= <i>produce, ready_meals</i> pred= $\emptyset$
h07	hard	Dinner my vegetarian sister and my meat-loving dad will both eat	mission_type, product_categories	mission_type: gold= <i>party_hosting</i> pred= <i>unclear</i> product_categories: gold= <i>produce,</i> <i>pantry_staples</i> pred= $\emptyset$
h08	hard	Just need something, I don't know, for later	risk_flags	risk_flags: gold= <i>ambiguous_request,</i> <i>missing_information</i> pred= <i>ambiguous_request</i>
h09	hard	Protein but no meat and no soy	dietary_constraints, product_categories	dietary_constraints: gold= <i>vegetarian</i> pred= $\emptyset$ product_categories: gold= <i>dairy_eggs,</i> <i>health_diet</i> pred= $\emptyset$
h11	hard	I'm diabetic, need a breakfast that won't spike my sugar	product_categories	product_categories: gold= <i>breakfast, dairy_eggs</i> pred= <i>breakfast</i>
h12	hard	Cheap but good quality wine for a dinner party	meal_occasion, risk_flags	meal_occasion: gold= <i>none</i> pred= <i>dinner</i> risk_flags: gold= <i>conflicting_goals</i> pred= $\emptyset$
h13	hard	Lactose-free milk and also some cheese for the kids	mission_type, preference_constraints, risk_flags	mission_type: gold= <i>kids_meal</i> pred= <i>household_stock_up</i> preference_constraints: gold= $\emptyset$ pred= <i>kid_friendly</i> risk_flags: gold= <i>ambiguous_request</i> pred= $\emptyset$
h15	hard	Vegan dinner, but my partner is allergic to nuts so be careful	mission_type, product_categories	mission_type: gold= <i>convenience_meal</i> pred= <i>health_goal</i> product_categories: gold= <i>produce, health_diet</i> pred= $\emptyset$

ID	Tier	Request	Failed Fields	Key difference
h16	hard	Get the usual	risk_flags	risk_flags: gold= <i>ambiguous_request</i> , <i>missing_information</i> pred= <i>ambiguous_request</i>